# PG No. 37 $D_{3h}(1)$ $\bar{6}m2$ (-62m setting) [ hexagonal ]

Table 1: Wyckoff site: `1o`, site symmetry: `-62m`

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | `[1,2,3,4,5,6,7,8,9,10,11,12]` |

Table 2: Wyckoff site: `2a`, site symmetry: `3.m`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, z]$ | `[1,2,3,10,11,12]` |
| 2 | $[0, 0, -z]$ | `[4,5,6,7,8,9]` |

Table 3: Wyckoff site: `3b`, site symmetry: `m2m`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, 0, 0]$ | `[1,4,8,11]` |
| 2 | $[0, x, 0]$ | `[2,5,7,10]` |
| 3 | $[-x, -x, 0]$ | `[3,6,9,12]` |

Table 4: Wyckoff site: `6c`, site symmetry: `..m`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, 0, z]$ | `[1,11]` |
| 2 | $[0, x, z]$ | `[2,10]` |
| 3 | $[-x, -x, z]$ | `[3,12]` |
| 4 | $[x, 0, -z]$ | `[4,8]` |
| 5 | $[0, x, -z]$ | `[5,7]` |
| 6 | $[-x, -x, -z]$ | `[6,9]` |

Table 5: Wyckoff site: `6d`, site symmetry: `m..`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, y, 0]$ | `[1,4]` |
| 2 | $[-y, x-y, 0]$ | `[2,5]` |
| 3 | $[-x+y, -x, 0]$ | `[3,6]` |
| 4 | $[y, x, 0]$ | `[7,10]` |
| 5 | $[x-y, -y, 0]$ | `[8,11]` |
| 6 | $[-x, -x+y, 0]$ | `[9,12]` |

Table 6: Wyckoff site: `12e`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, y, z]$ | `[1]` |

*continued ...*

Table 6

| No. | position | mapping |
|---|---|---|
| 2 | $[-y,\, x-y,\, z]$ | [2] |
| 3 | $[-x+y,\, -x,\, z]$ | [3] |
| 4 | $[x,\, y,\, -z]$ | [4] |
| 5 | $[-y,\, x-y,\, -z]$ | [5] |
| 6 | $[-x+y,\, -x,\, -z]$ | [6] |
| 7 | $[y,\, x,\, -z]$ | [7] |
| 8 | $[x-y,\, -y,\, -z]$ | [8] |
| 9 | $[-x,\, -x+y,\, -z]$ | [9] |
| 10 | $[y,\, x,\, z]$ | [10] |
| 11 | $[x-y,\, -y,\, z]$ | [11] |
| 12 | $[-x,\, -x+y,\, z]$ | [12] |